Contents

1 Appendix E: Spectral Unmixing and Classification

1.1 Objective

Comprehensive spectral analysis: realistic CHC data generation, feature extraction, unmixing (NMF, VCA, ICA), and multi-algorithm classification with deterministic evaluation.

1.2 Methods (src)

- `src/case_studies/spectral_unmixing.py`
  - `generate_realistic_chc_spectra(n_compounds: int, n_wavelengths: int, seed: int=42) -` — synthetic CHC spectra with ground truth
  - `nmf_unmix(spectra: np.ndarray, n_components: int, seed: int=42) -> (W, H)` — deterministic NMF
  - `vertex_component_analysis(spectra: np.ndarray, n_endmembers: int) -> np.ndarray` — VCA endmember extraction
  - `independent_component_analysis_spectra(spectra: np.ndarray, n_components: int) -> np` — ICA separation
  - `spectral_feature_extraction(spectra: np.ndarray, wavelengths: np.ndarray, method: st` — peaks, derivatives, PCA, statistical features
  - `advanced_classification_suite(features: np.ndarray, labels: np.ndarray) -> dict` — multi-algorithm benchmark
  - `performance_metrics_comprehensive(y_true: np.ndarray, y_pred: np.ndarray, y_prob: Op`
  - `lda_baseline(features: np.ndarray, labels: np.ndarray, seed: int=42) -> dict` — closed-form LDA baseline

1.3 Script and outputs

- Script: `scripts/generate_spectral_unmixing.py`
- Data: `output/data/spectral_unmixing_comprehensive.npz`
- Figure: `figures/spectral_unmixing_comprehensive_analysis.png`

1.4 Figure

1.5 Equation References

- Spectral overlap: see (**??**) analogs for information metrics; overlap in main text.

[width=0.9]figures/spectral$_unmixing_comprehensive_analysis.png$

Figure 1: *Comprehensive spectral analysis generated deterministically: synthetic CHC spectra, NMF/VCA/ICA unmixing, multi-method feature extraction, and classification benchmarks.*

[width=0.9]figures/spectral$_unmixing_comprehensive_analysis.png$

Figure 2: *Spectral unmixing and classification results produced by* ‘$scripts/generate_spectral_unmixing.py$‘$using$‘$src/case_studies/spectral_unmixing.py$‘$.Panelsshowmixedspectra, recovered$

1.6 Reproducibility

- Run: `python3 scripts/generate_spectral_unmixing.py`
- Artifacts saved to `output/data/` and `figures/`.
- Fixed RNG seed (42) used for deterministic NMF initialization and cross-validation splits.

1.7 Cross-references

- Methods: sec:methodology
- Symbols: sec:symbols$_glossaryMathappendix : sec : mathematical_appendix$

[width=0.9]figures/integrated$_a nalysis_c ross_d omain_s ynthesis.png$

Figure 3: *Integrated classification benchmark from* ‘$scripts/generate_i ntegrated_a nalysis.py$‘$summarizingclassificationperformanceacrossspectralandneuralfeaturesets.$